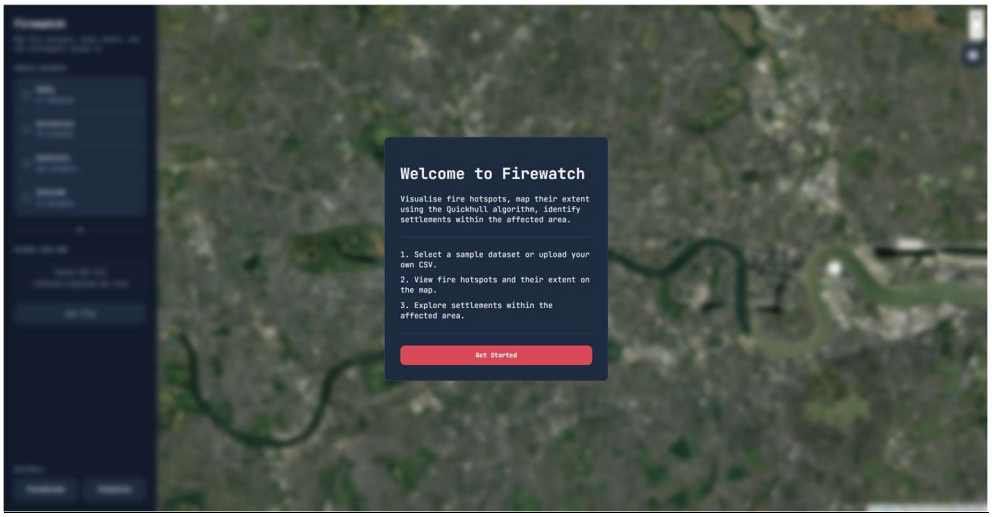
End To End Testing

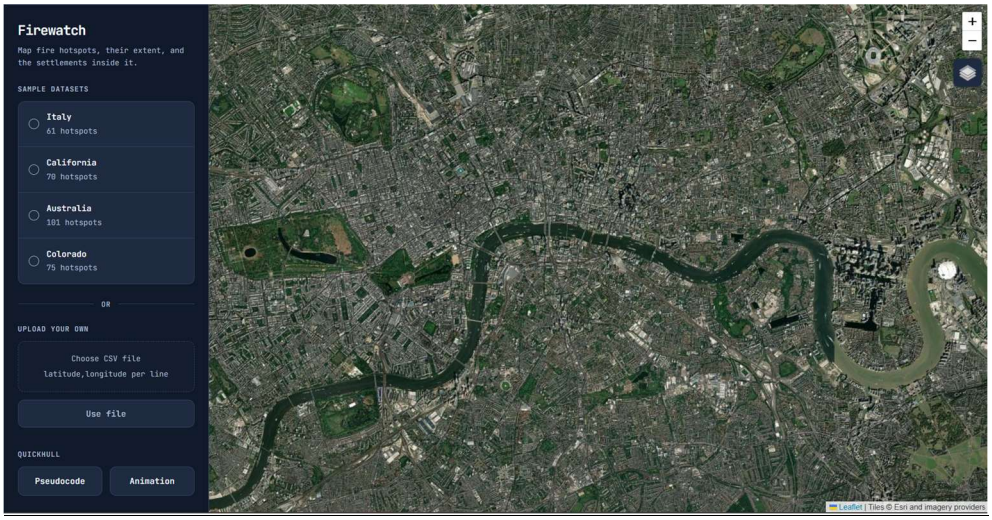
Map Display

Test World Map Loads Upon Program Start

- Expected Outcome: The world map is visible after the welcome message has been accepted
- Action Taken: Dismissed website entry pop-up
- Result: Success

Evidence:

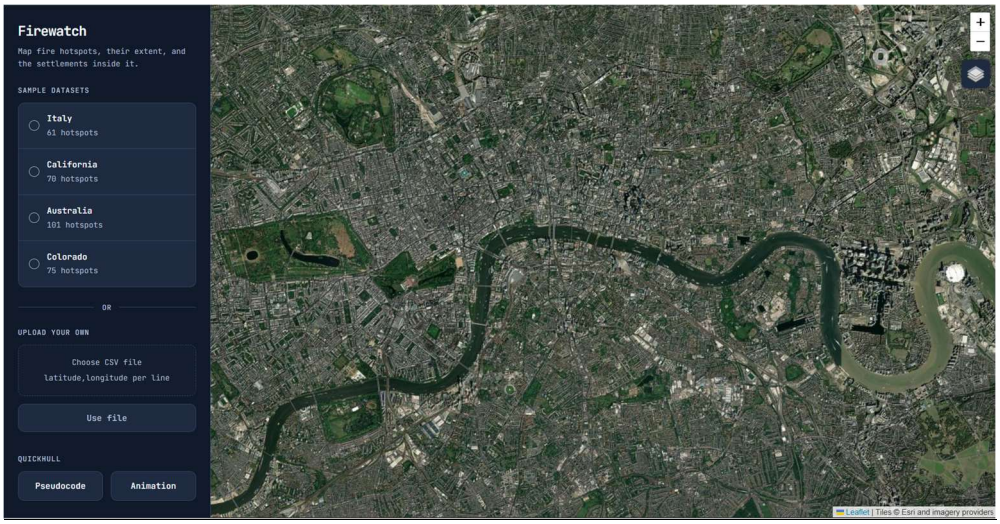




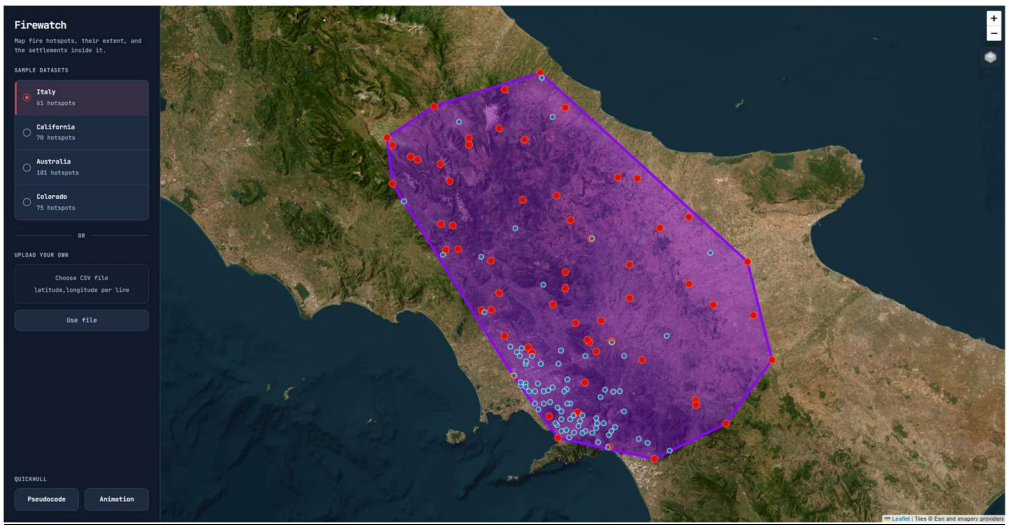
Test World Map Zoom Out Function

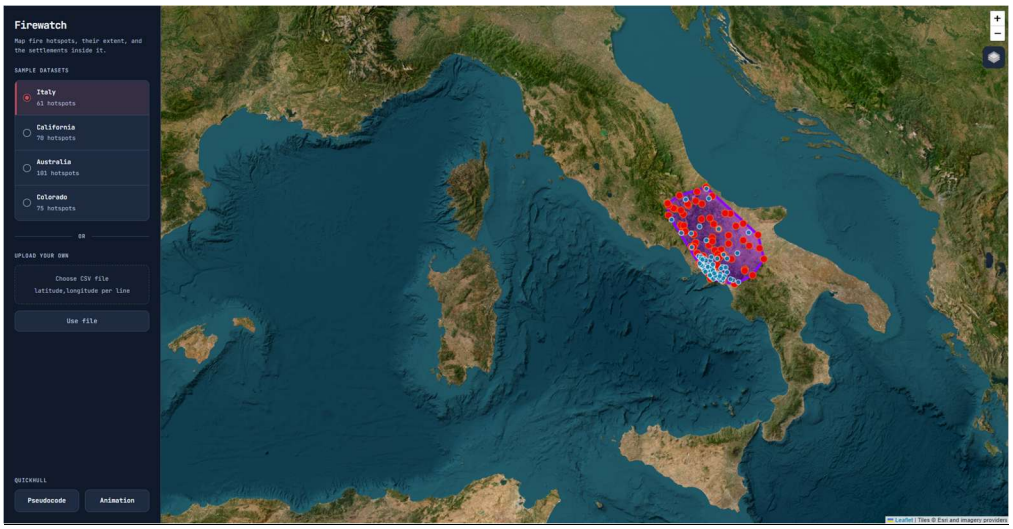
- Expected Outcome: When scrolling down on the mouse wheel or clicking the `-` button, the world map will zoom out a set amount, with the active hull scaling accordingly and taking up less of the screen space.
- Action Taken: Mouse wheel scroll and `-` button click
- Result: Success

Evidence:


Firewatch
Map fire hotspots, their extent, and the settlements inside it.
SAMPLE DATASETS
Italy
61 hotspots
California
70 hotspots
Australia
101 hotspots
Colorado
75 hotspots
OR
UPLOAD YOUR OWN
Choose CSV file
latitude,longitude per line
Use file
QUICKHULL
Pseudocode
Animation
Leaflet | Tiles © Esri and imagery providers


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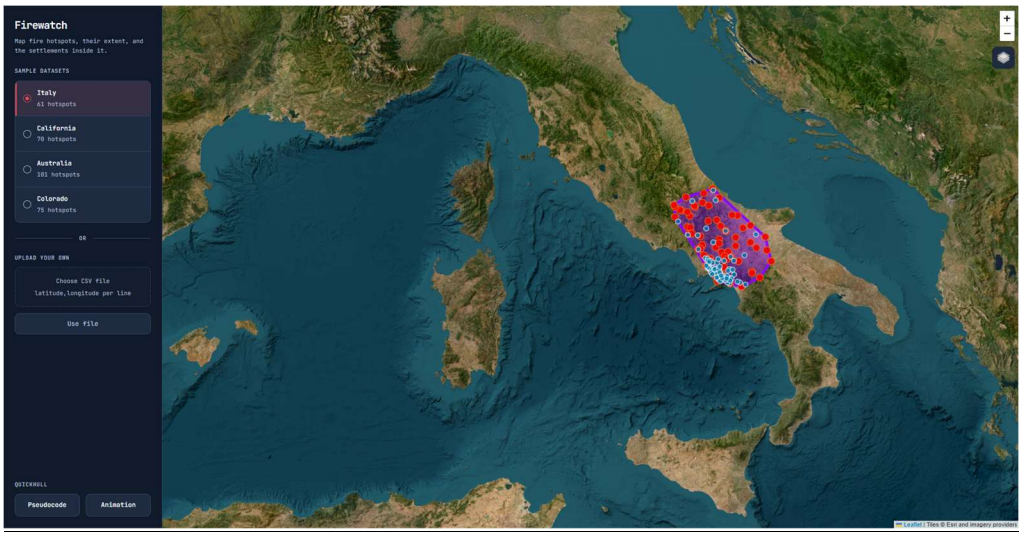
Test World Map Zoom In Function

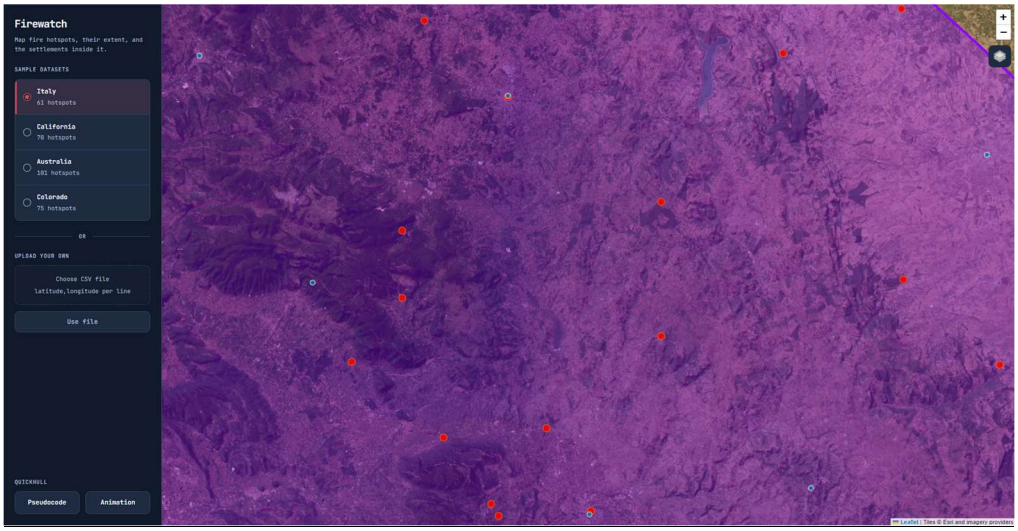
- Expected Outcome: When scrolling up on the mouse wheel or clicking the `+` button, the world map will zoom in a set amount, with the active hull scaling accordingly and taking up more of the screen space.
- Action Taken: Mouse wheel scroll and `+` button click
- Result: Success

Evidence:





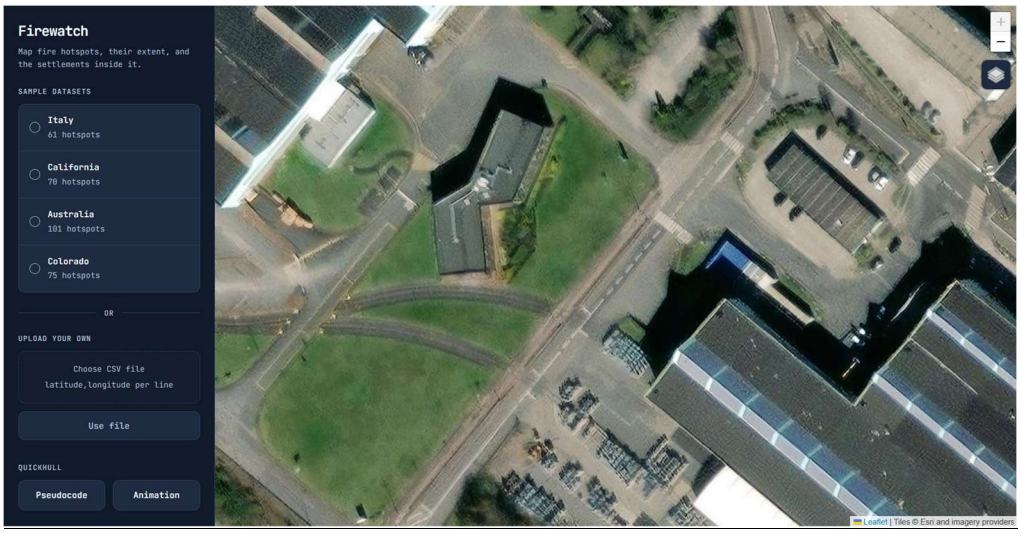




[Test World Map Zoom In To Max]

- Expected Outcome: After a certain amount of zooming in, further attempts to zoom in will yield no changes, and a granular level of detail should be visible. Zooming in on the `terrain` display setting ought to change the geography and place names displayed.
- Action Taken: Continuously zooming in on both the terrain and satellite modes.
- Result: Failure, when zooming in to max with `Terrain` view, an ugly error message was displayed.

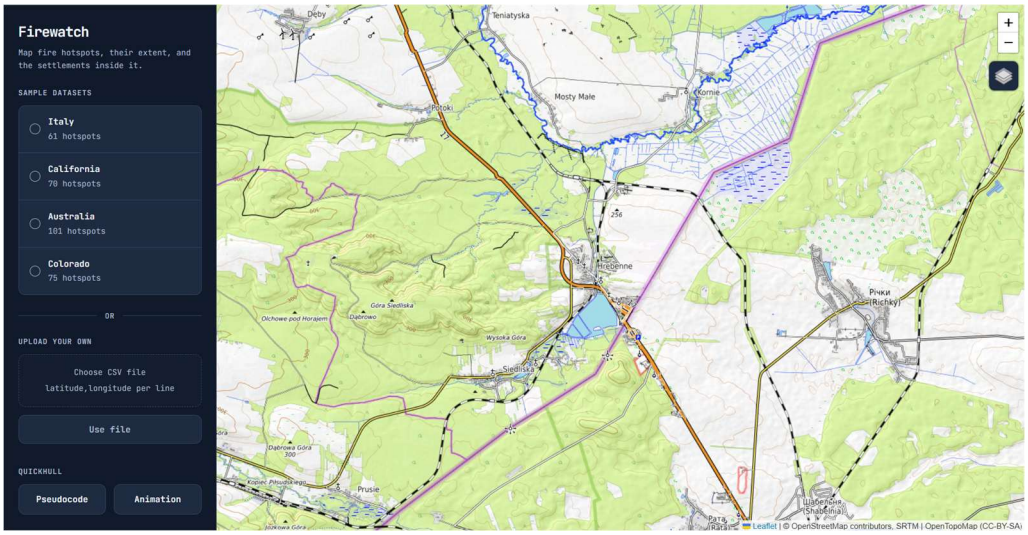
Evidence:



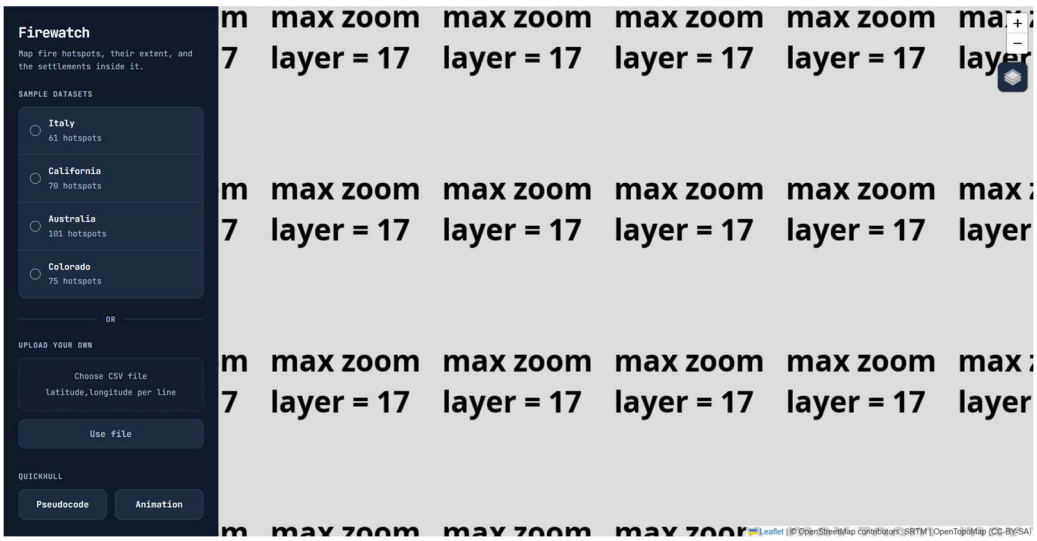



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UPLOAD YOUR OWN
Choose CSV file
latitude,longitude per line
Use file
QUICKHULL
Pseudocode
Animation
Leaflet | © OpenStreetMap contributors, SRTM | OpenTopoMap (CC-BY-SA)


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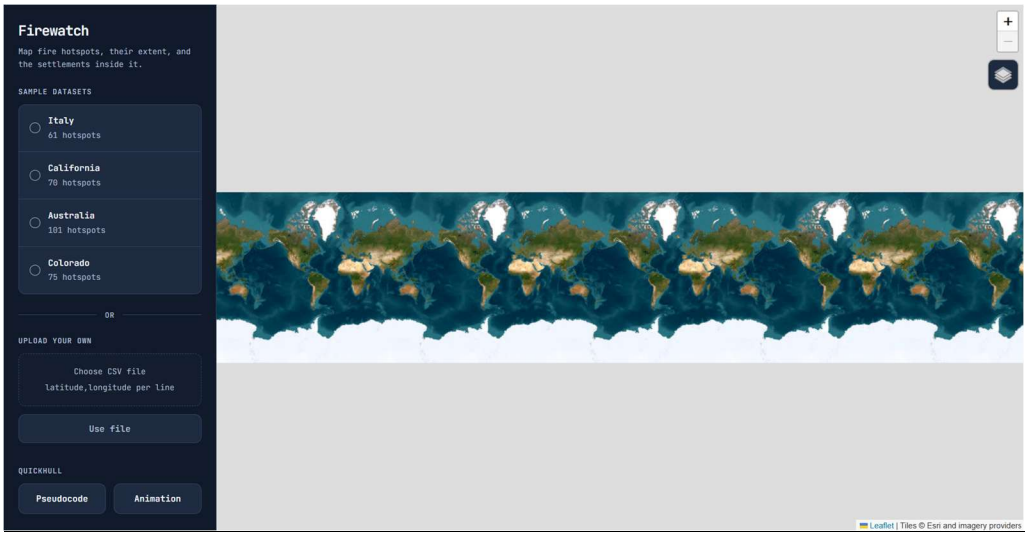




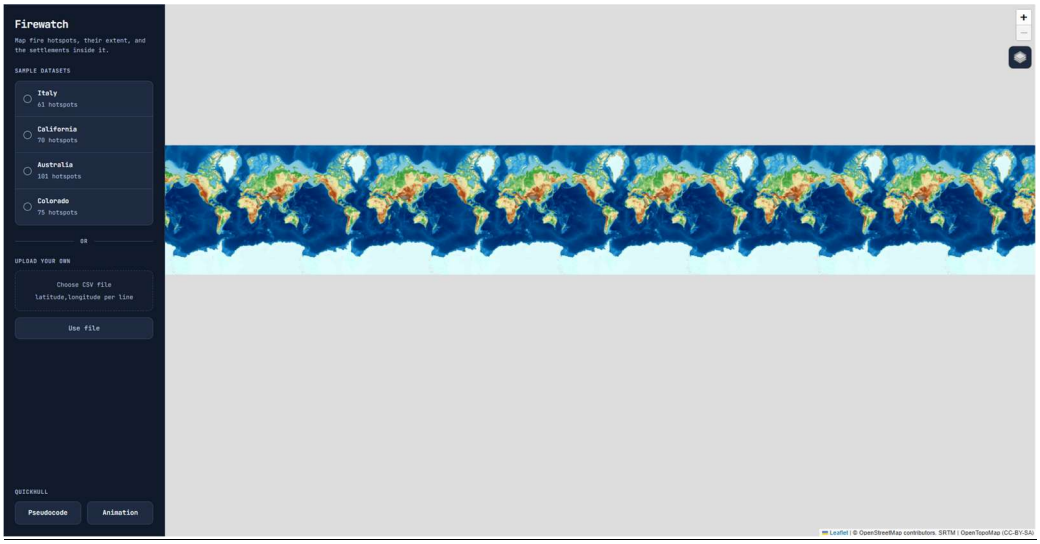
Test Zoom Out To Max

- Expected Outcome: After a certain amount of zooming out, further attempts to zoom out will yield no changes, and multiple iterations of the 2D world map should be displayed. This applies to both view modes.
- Action Taken: Continuously zooming out on both the terrain and satellite modes.
- Result: Success

Evidence:







[Test Scroll World Map]

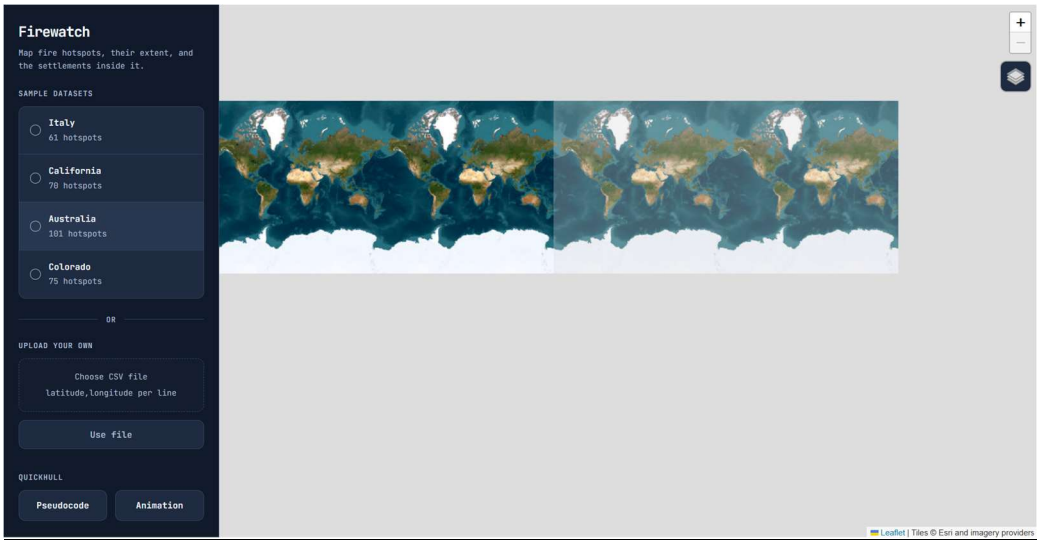
- Expected Outcome: Dragging the world map with the mouse should move the display to a different area of the world. Continuously scrolling beyond the horizontal borders of world should cause a copy of the world to display ad infinitum
- Action Taken: Dragging the mouse across the screen vertically and horizontally
- Result: Success

Evidence:









Hull

Test Hull Preparation

- Expected Outcome: When loading a convex hull, the map should reposition to the location of the hull and zoom to an appropriate scale; whereby the entire hull is visible, and its shape and points can be clearly made out. A loading wheel and message should be displayed
- Action Taken: Clicked on a sample dataset to load it
- Result: Success

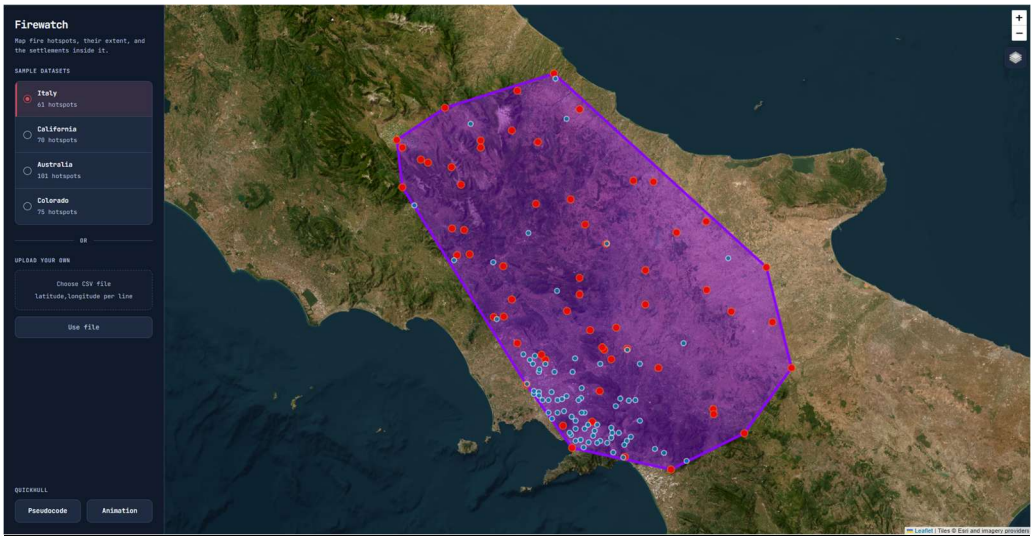
Evidence:

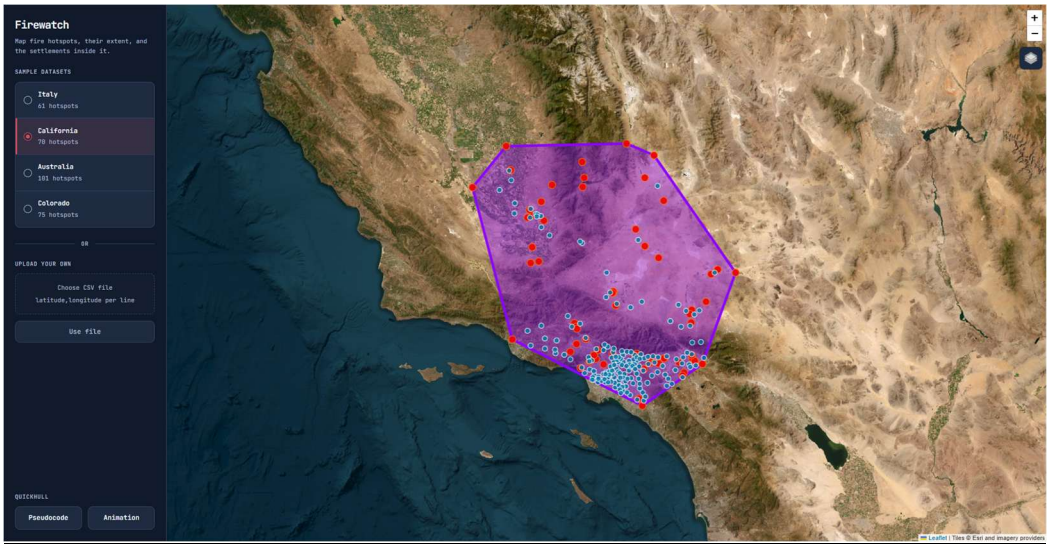


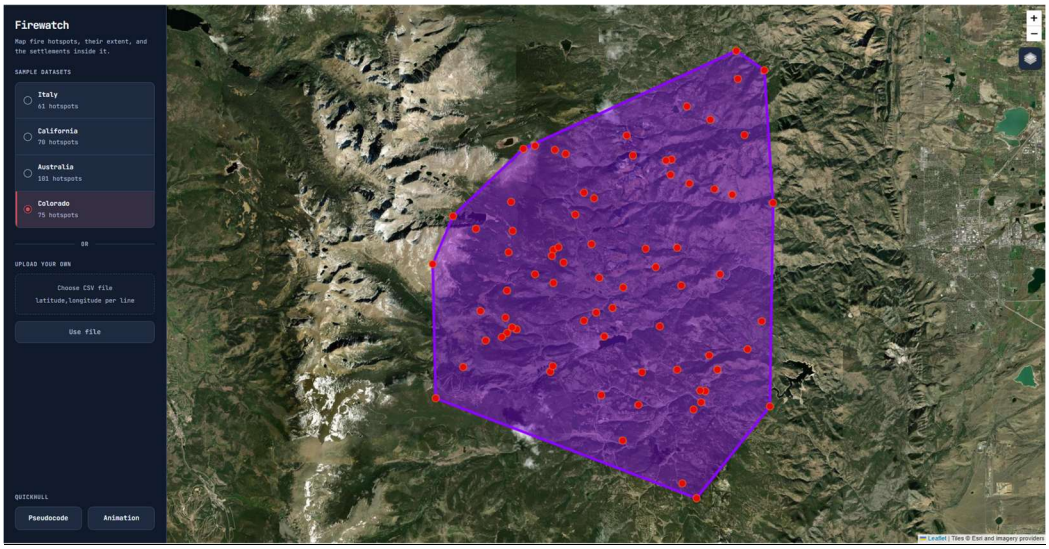
Test Generates Convex Hull Correctly From Samples

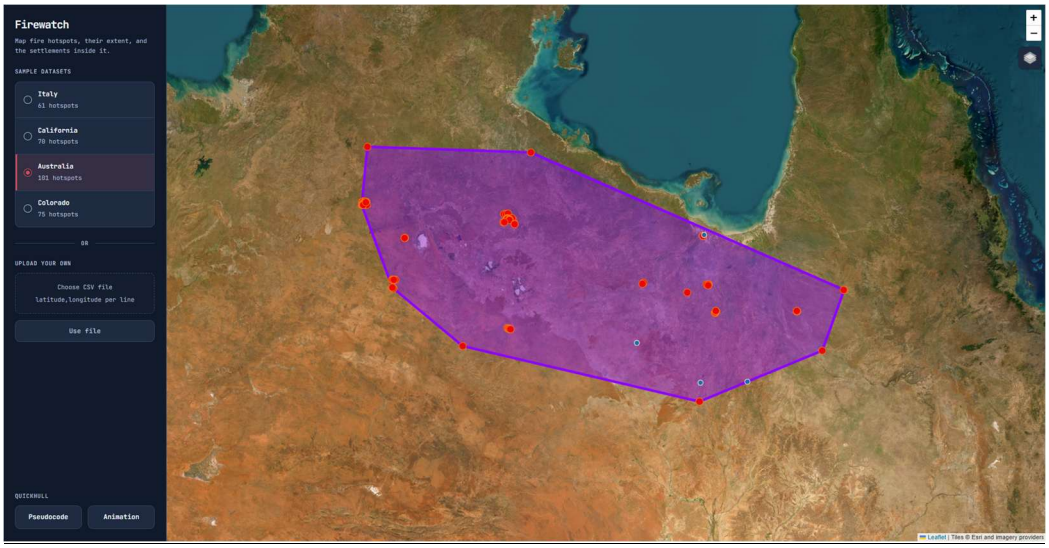
- Expected Outcome: Each sample should produce a corresponding hull featuring the correct number of fire co-ordinates represented as red points, with a convex hull displayed whereby all exterior points are connected by a line, and the area within is shaded purple. Settlements within the hull should appear as blue dots.
- Action Taken: Clicked each of the sample datasets consecutively
- Result: Success

Evidence:





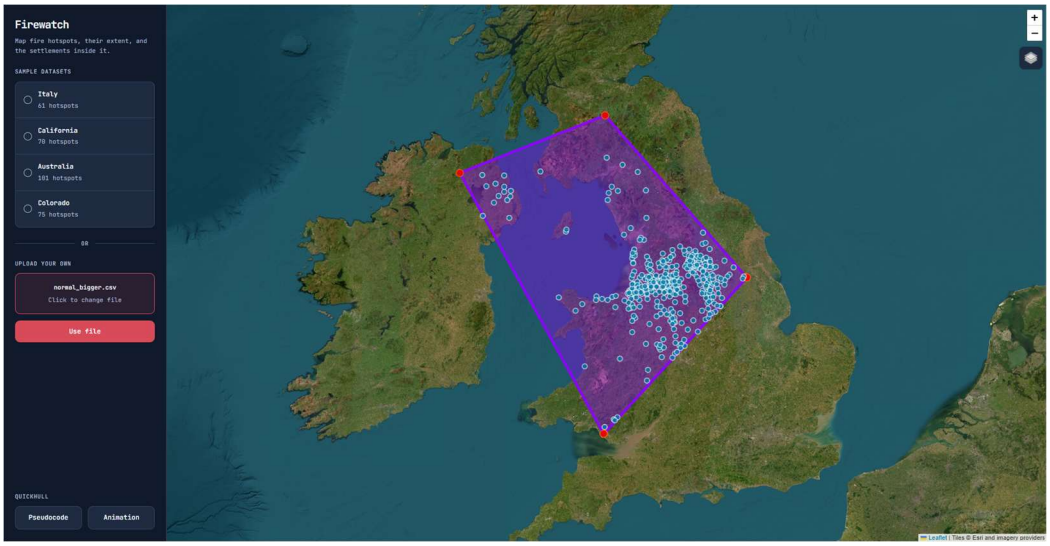


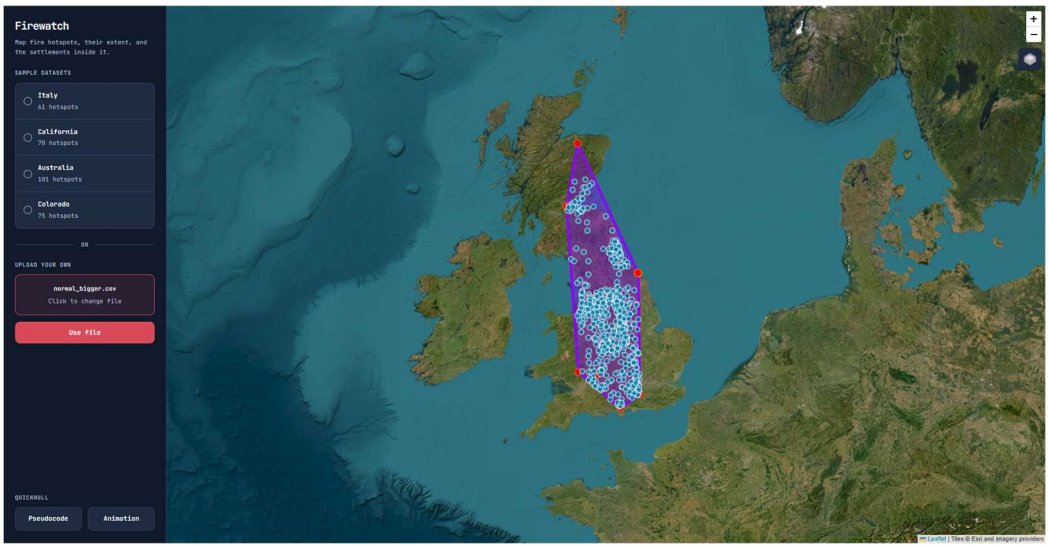


[Test Generates Convex Hull From CSV]

- Expected Outcome: Generating a hull from a valid CSV file of co-ordinates sample should produce a valid convex hull featuring the correct number of fire co-ordinates represented as red points, with a convex hull displayed whereby all exterior points are connected by a line, and the area within is shaded purple. Settlements within the hull should appear as blue dots. Uploading a second CSV file in place of the original and generating a hull again should produce a new convex hull corresponding to the points within the new file.
- Action Taken: Uploaded valid CSV file and clicked `Use file` button
- Result: Success

Evidence:





Test Renders CSV Hull With Out Of Bounds Data

- Expected Outcome: When rendering a hull from a CSV which contains out of bounds data (e.g. latitude=120.0), the hull and its points should still be

displayed, but out of bounds lines should be ignored for both the hull edges and interior points.

- Action Taken: Uploaded CSV file with out of bounds data and clicked `Use file` button
- Result: Failure, out of bounds point was plotted



CSV Upload

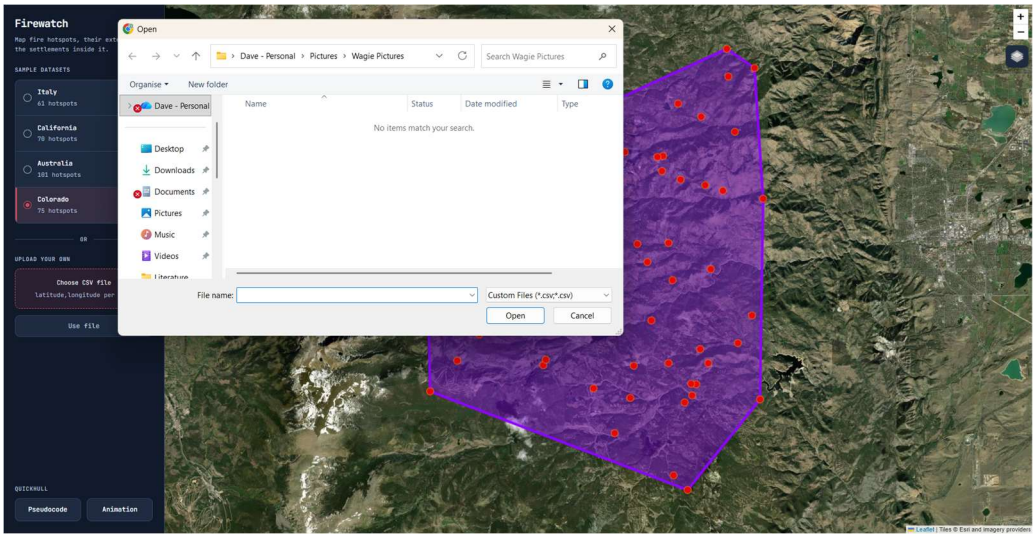
Test Upload Valid CSV file

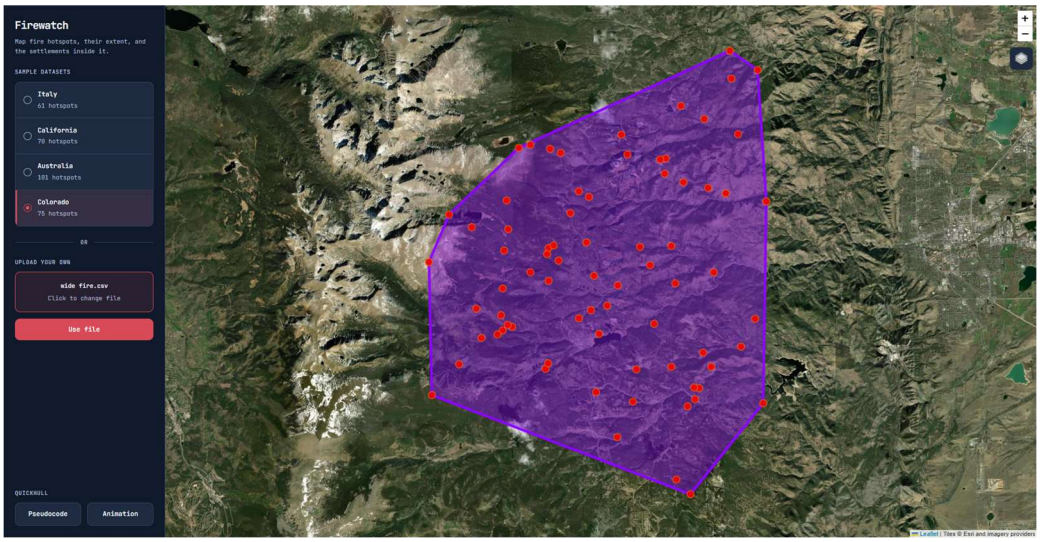
- Expected Outcome: Hovering over the `Choose CSV file` box or ` should highlight it. Clicking the `Choose CSV file` box should bring up a file selection option using the user's directories. Once a file has been selected, it should appear in the CSV file box.
- Action Taken: Hovered mouse over `Choose CSV file` box. Clicked `Choose CSV file` and selected a valid CSV file from a folder to upload
- Result: Success

Evidence:

Test Hover Over CSV UI Components







[Test Can’t Create Hull From CSV When No File Has Been Uploaded]

- Expected Outcome: Hovering over the `Use file` button when no CSV has been uploaded should not highlight it. Clicking the same button should produce no effect.
- Action Taken: Hovered over `Use file` with mouse and clicked it.
- Result: Success

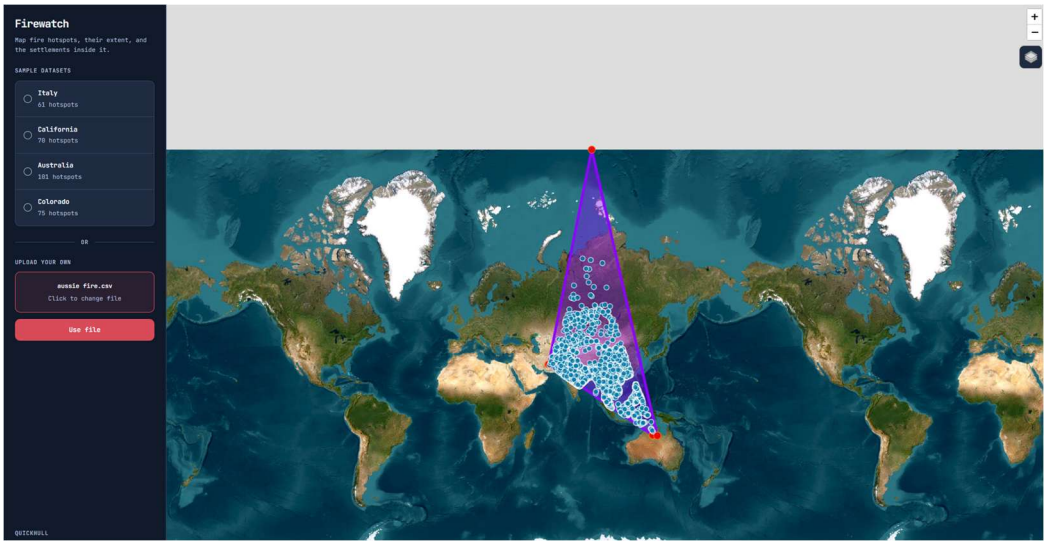


Test Creating Large Hull From Csv Zooms Out

- Expected Outcome: When creating a hull from a set of points spanning a large area, the map will zoom out to display the entire hull.
- Action Taken: Uploaded wide area hull CSV and clicked `Use file`
- Result: Success

Evidence:





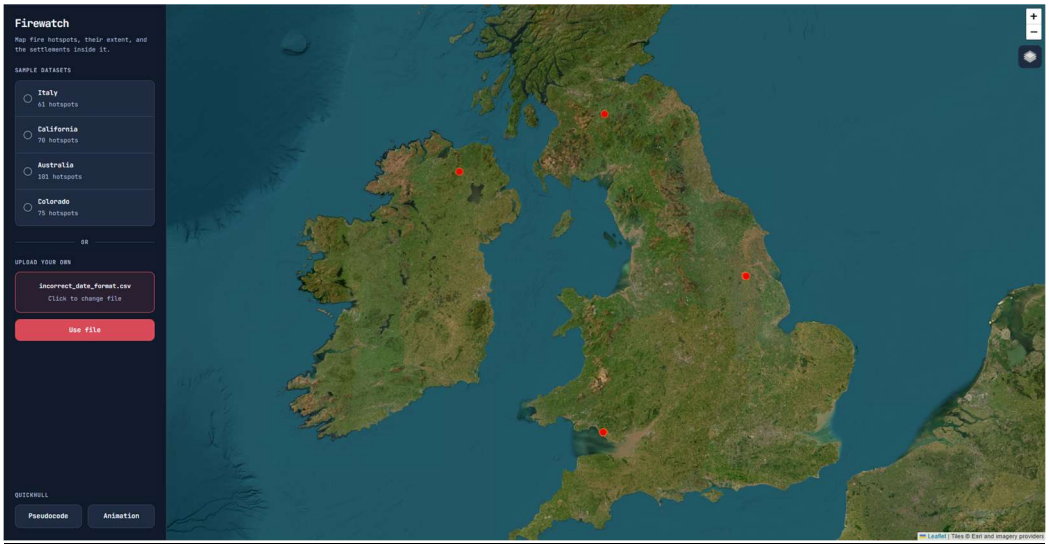
Test CSV Upload Rejects Invalid CSV With Error message

- Expected Outcome: When attempting to create a hull using a CSV containing an invalid set of points, an error message should be displayed. Clicking the button on the error message should cause it to disappear, and a hull should not appear
- Action Taken: Uploaded an invalid CSV, clicked `Use file`, then clicked `I understand`.
- Result: Success

Evidence:





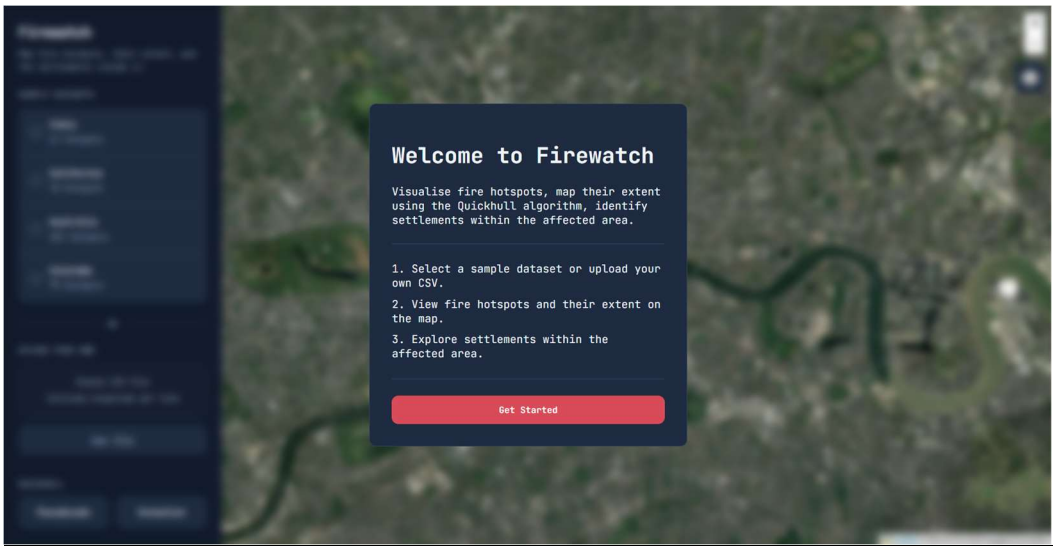


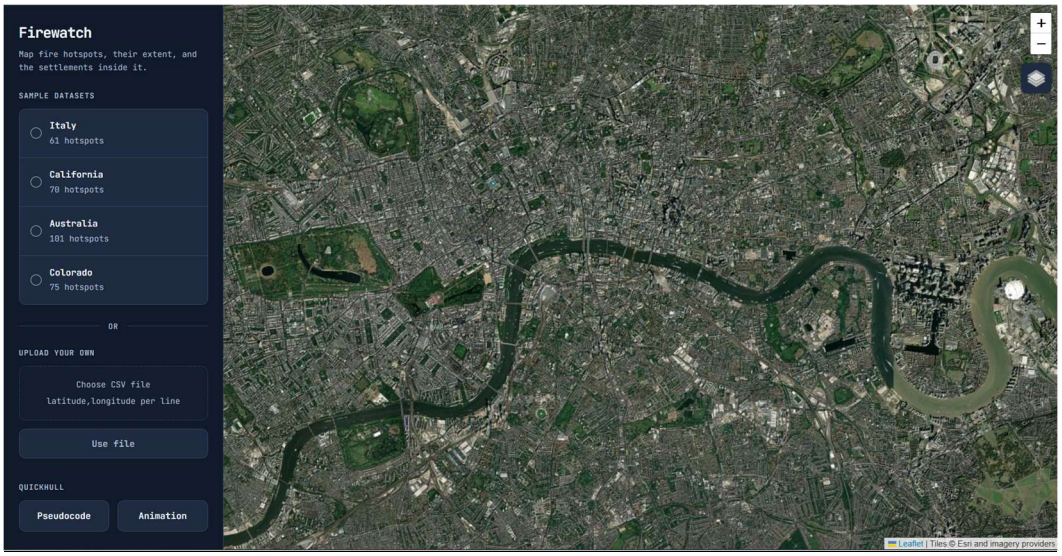
UI Interactivity

Test Website Displays Pop-up Message Upon Entry

- Expected Outcome: Upon first entering the website, a welcome message should be displayed
- Action Taken: Opened website
- Result: Success

Evidence:

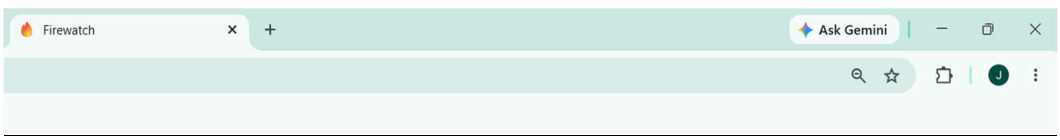




Test Firewatch Tab Display

- Expected Outcome: The tab for the website should display the text `FireWatch` next to a flame symbol.
- Action Taken: Opened website
- Result: Success

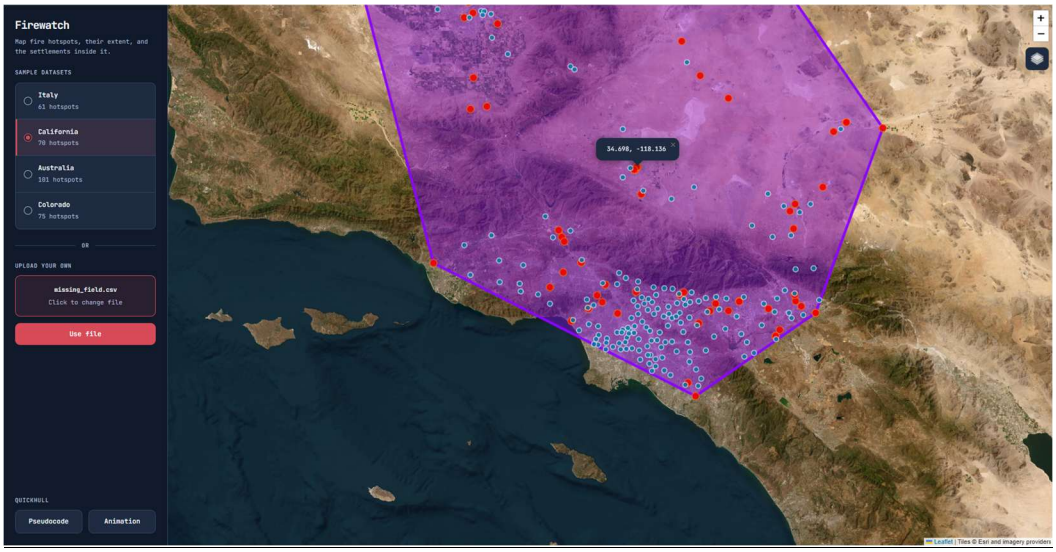
Evidence:

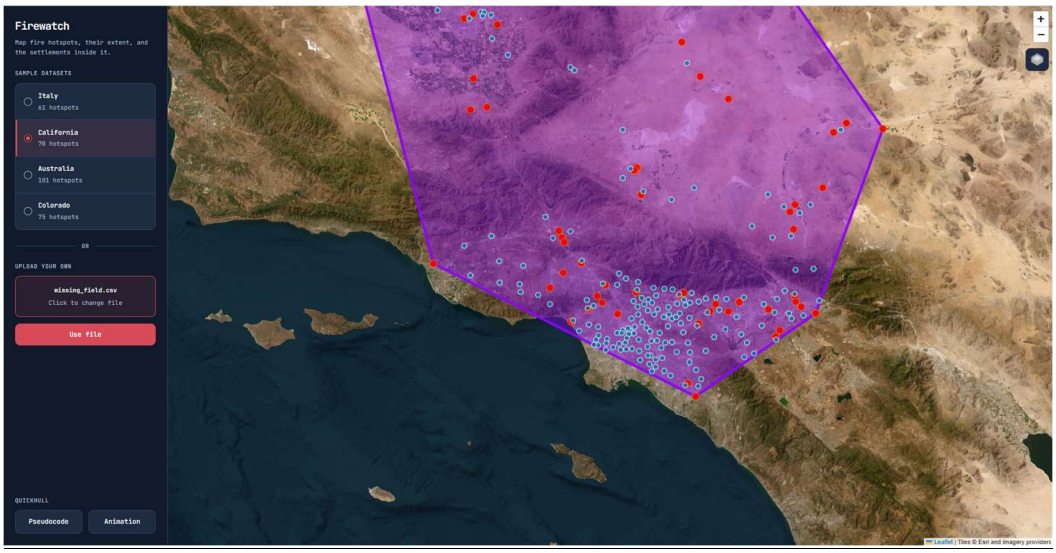


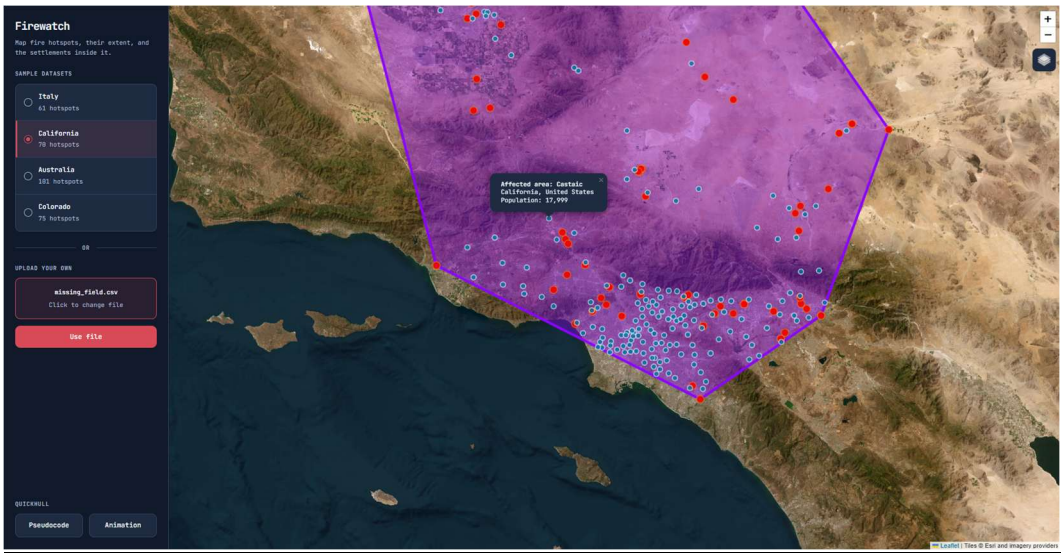
Test Clicking A Point Displays Information

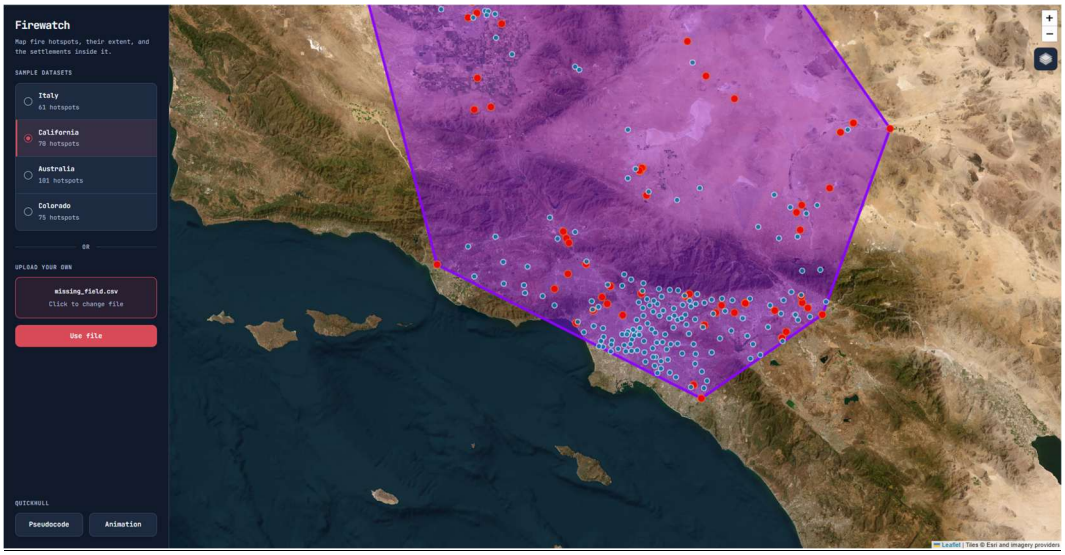
- Expected Outcome: Clicking a red point ought to raise a bubble showing the co-ordinate of that point. Clicking a blue point should raise a bubble naming the affected settlement, its population and its country. Clicking the x in the corner of a bubble or clicking elsewhere on the screen should dismiss it.
- Action Taken: Clicked red point, clicked `x` on subsequent pop-up, clicked blue point, then clicked a different point on the screen.
- Result: Success

Evidence:







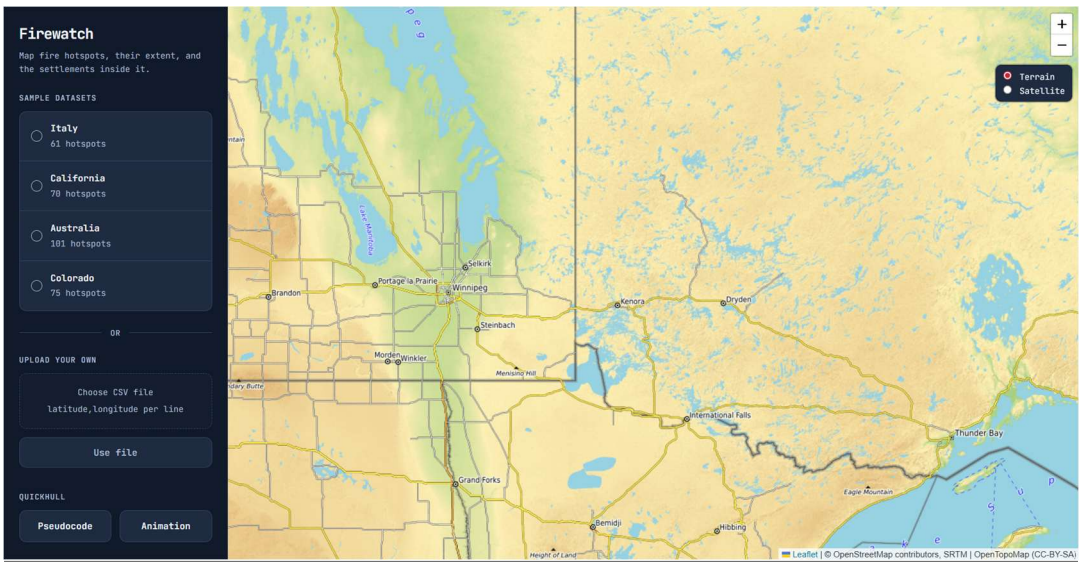


Test Map Type Change

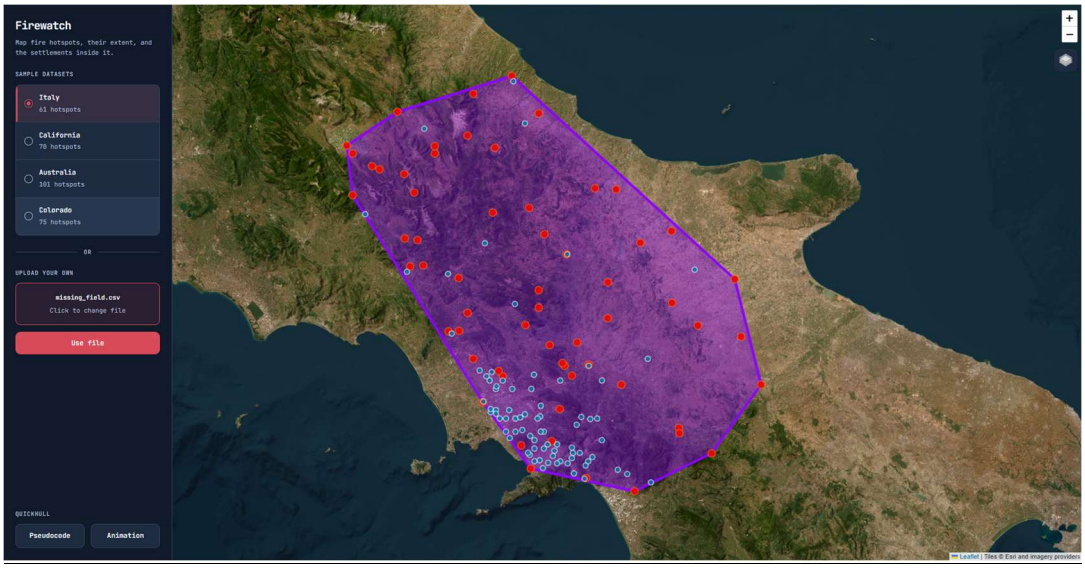
- Expected Outcome: Hovering over the `Layer` symbol should bring up a drop-down list with `Satellite` and `Terrain` options. Selecting `Terrain` while on satellite display should change the display type to terrain, and vice versa. Changing display type should not affect the hull presently on screen if there is one.
- Action Taken: Hovered mouse over layer symbol and clicked `Terrain`, then `Satellite`. Followed the same procedure with an active hull.
- Result: Success

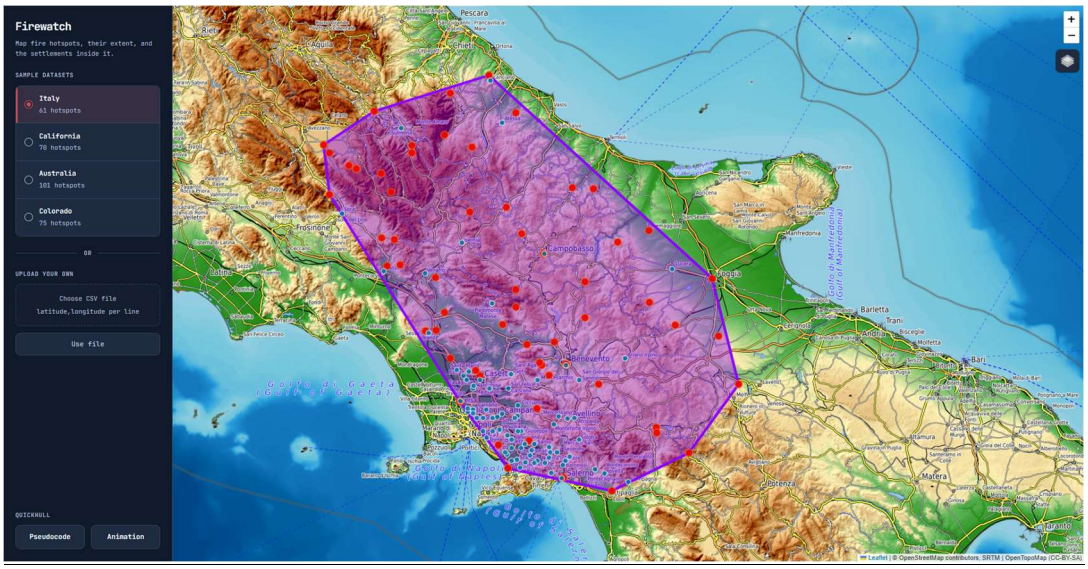
Evidence:

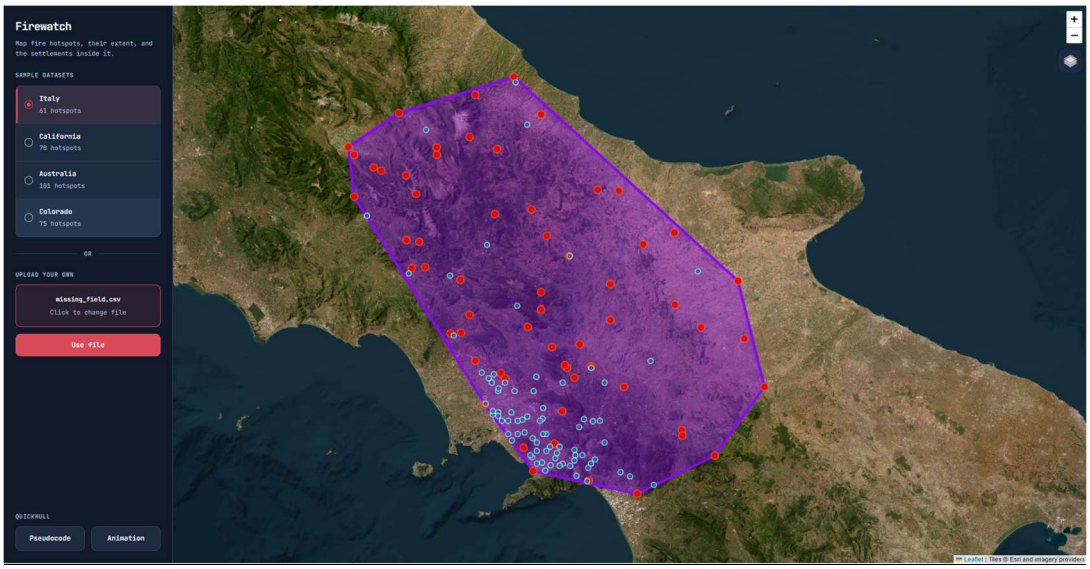

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Map fire hotspots, their extent, and the settlements inside it.
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Italy
61 hotspots
California
70 hotspots
Australia
101 hotspots
Colorado
75 hotspots
OR
UPLOAD YOUR OWN
Choose CSV file
latitude,longitude per line
Use file
QUICKHULL
Pseudocode
Animation
Terrain
Satellite
Leaflet | Tiles © Esri and imagery providers


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Satellite
Selkirk
Portage la Prairie
Winnipeg
Brandon
Steinbach
Morden
Winkler
Kenora
Dryden
International Falls
Thunder Bay
Grand Forks
Bemidji
Hibbing
Eagle Mountains
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missing_field.csv
Click to change file
Use file
QUICKHULL
Pseudocode
Animation
Leaflet | Tiles © Esri and imagery providers

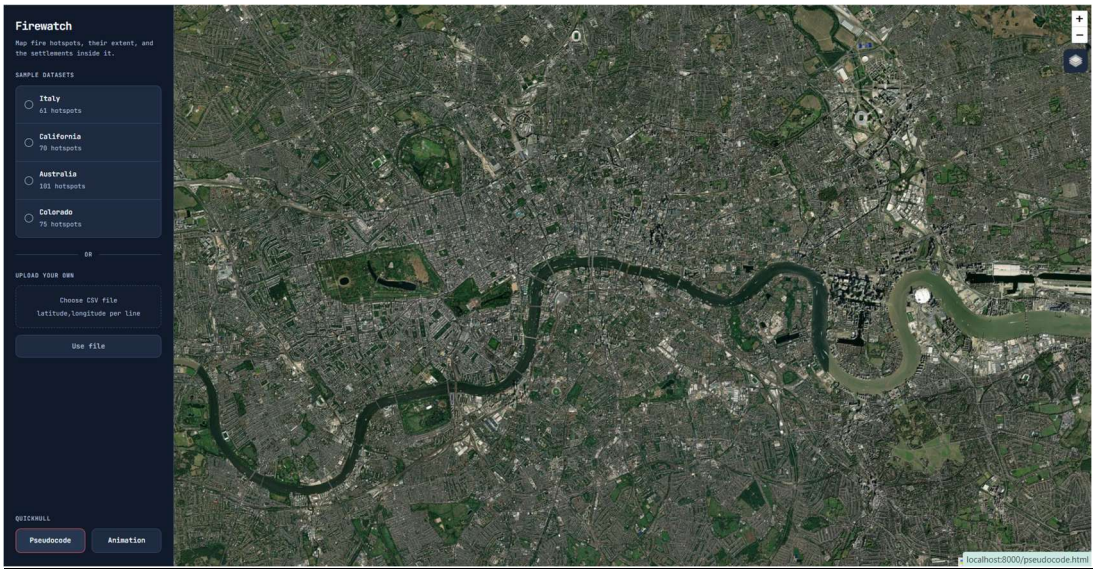

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Pescara
Campobasso
Benevento
Barletta
Trani
Bari
Matera
Potenza
Taranto
Leaflet | © OpenStreetMap contributors, SRTM | OpenTopoMap (CC-BY-SA)

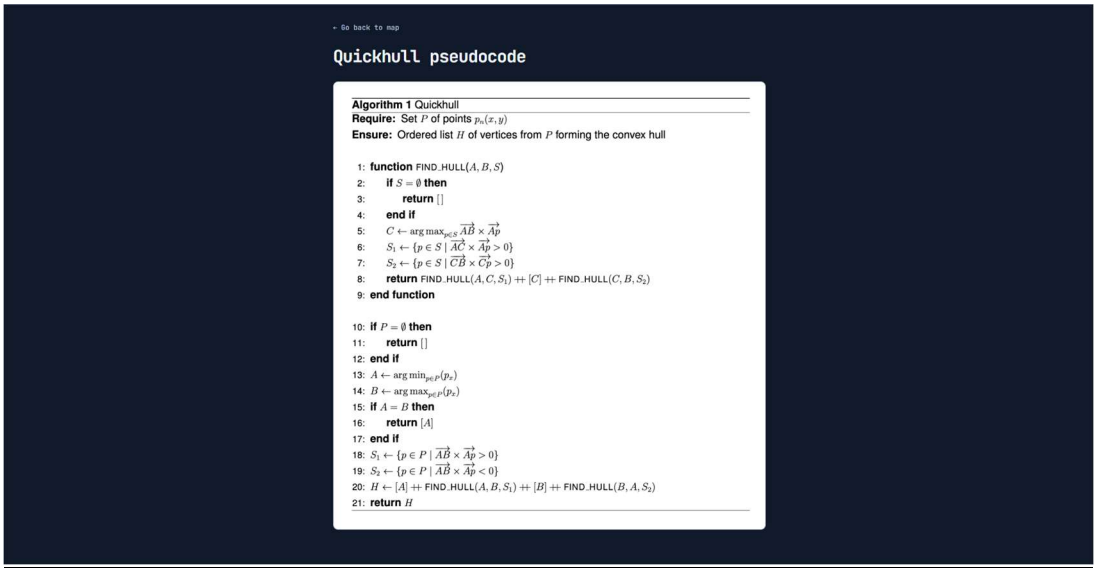

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Animation
Leaflet | Tiles © Esri and imagery providers

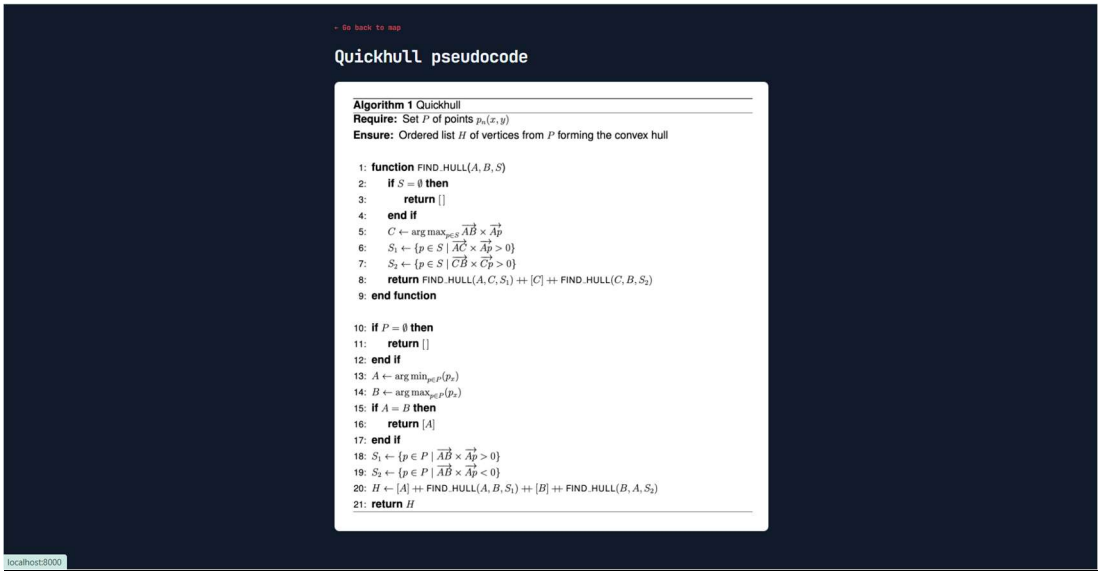
Test Quickhull Pseudocode Navigation

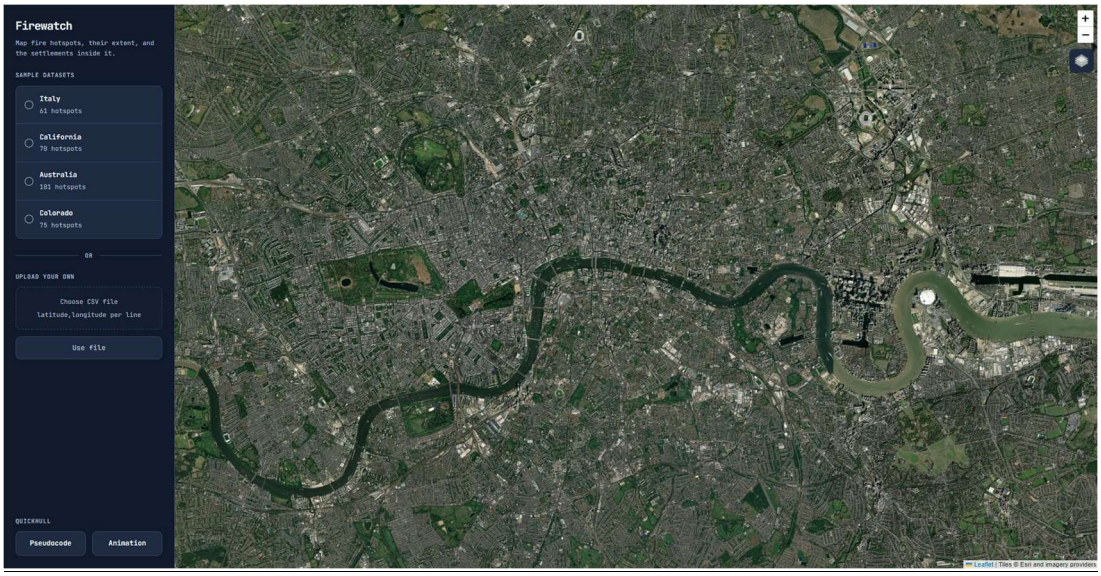
- Expected Outcome: Hovering over the `Pseudocode` button will highlight it, and clicking on it will transport the user to the Pseudocode page. Clicking on the `Go back to map` text on this page will navigate back to the map display
- Action Taken: Hovered over the `Pseudocode button` and clicked it. Clicked on `Go back to map` on the subsequent page.
- Result: Success

Evidence:









Test Animation

- Expected Outcome:
- Action Taken:
- Result:

Evidence:


Firewatch
Map fire hotspots, their extent, and the settlements inside it.
SAMPLE DATASETS
Italy
61 hotspots
California
76 hotspots
Australia
101 hotspots
Colorado
75 hotspots
OR
UPLOAD YOUR OWN
Choose CSV file
latitude,longitude per line
Use file
QUICKHULL
Pseudocode
Animation
+
−
localhost:8000/animation.html